#Aviation Meteorology

Chapter Four: Reduced Surface Visibility

Objectives

- To Define fog and Identify different types of fog
- To Know The relationship of visibility to fog type and duration
- To be familiarized with Other weather phenomena causing reduced surface visibility

Reduced Visibility

- Reduced surface visibility refers to a condition where the visibility on the ground or water is significantly decreased due to various *atmospheric factors*.
- It can be caused by weather conditions such as *fog, mist, smoke, haze, or dust, as well as by human activities such as construction or pollution*.

- Reduced surface visibility can be hazardous for transportation, as it can make it difficult for drivers, pilots, and navigators to see and avoid obstacles.
- In extreme cases, it can even lead to accidents and fatalities.

- To avoid accidents due to reduced surface visibility, it's important to take appropriate precautions.
- Pilots and navigators should rely on their instruments and follow procedures for flying in reduced visibility conditions.

- In general, it's important to stay informed about weather and air quality conditions in your area and to take appropriate measures to ensure your safety and the safety of others when visibility is reduced.

Fog

- Fog is a cloud that forms near the ground, reducing visibility to less than 1 km.
- It is typically composed of tiny water droplets or ice crystals that are suspended in the air.
- There are several types of fog, and each type forms and dissipates in a different way.

- Here are the most common types of fog and how they form and dissipate:

- **Radiation Fog**: Radiation fog forms on ***clear, calm nights*** when the ground *cools rapidly by radiation*.
  - ✓ This causes the air near the ground to ***cool***, and if the temperature drops ***to the dew point***, water droplets form, creating fog.
  - ✓ Radiation fog is typically thickest near sunrise and can dissipate when the sun warms the ground, or if the wind picks up.

- **Advection Fog**: Advection fog forms when *warm, moist air moves over a cold surface*, such as snow or water.
  - ✓ As the warm air cools, water droplets form, creating fog.
  - ✓ Advection fog can persist for a long time, especially if the air continues to move over the cold surface.
  - ✓ It can dissipate if the wind stops or if the air warms up.

- **Upslope Fog**: Upslope fog forms when ***moist air is forced up a hill or mountain slope***.
  - ✓ As the air rises, it cools, and if the temperature drops to the dew point, water droplets form, creating fog.
  - ✓ Upslope fog can persist as long as the air is forced up the slope.
  - ✓ It can dissipate if the air stops rising or if the temperature rises.

- **Steam Fog**: Steam fog, also known as sea smoke or evaporation fog, forms when *cold air moves over warm water*.
  - ✓ As the cold air cools the water, water vapor rises and condenses, creating fog.
  - ✓ Steam fog can dissipate if the air warms up or if the wind picks up.

- **Frontal Fog**: Frontal fog forms along the leading edge of a warm front as warm, moist air is lifted over cooler air.
  - ✓As the air rises, it cools and water droplets form, creating fog.
  - ✓Frontal fog can persist as long as the warm front continues to move over the cooler air.
  - ✓It can dissipate if the front moves through, or if the air warms up.
  - The key difference between Advection and frontal fog lies in the meteorological conditions that lead to their formation.
  - Advection fog is formed when warm, moist air moves horizontally over a *colder surface*, while frontal fog is associated with the lifting of warm, moist air over a front, often accompanied by precipitation.

- In general, fog dissipates when the *temperature rises* or the *wind picks up*, causing the fog droplets to evaporate or disperse.
- However, fog can persist for long periods of time, especially if the conditions that caused it to form persist.

Relationship Of Visibility To Fog Type And Duration

- Fog can significantly reduce visibility in aviation, posing a hazard to pilots and making it difficult to take off and land safely.
- The effects of each fog type on reducing visibility in aviation are as follows:
- **Radiation Fog**: Radiation fog can reduce visibility to less than 1/4 mile (400 meters), making it difficult for pilots to see runways, taxiways, and other aircraft.
  - ✓It can also cause instrument malfunctions and affect communication systems.

❖ **Advection Fog**: Advection fog can reduce visibility to less than 1 mile (1.609 km) and can persist for long periods of time, making it difficult for pilots to navigate and maintain visual contact with the ground.

- **Upslope Fog**: Upslope fog can reduce visibility to less than 1/4 mile (400 meters) and can persist for long periods of time, making it difficult for pilots to navigate and maintain visual contact with the ground.
  - ✓ It can also cause icing on the aircraft.

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- In general, fog can make flying more challenging and increase the risk of accidents.
- Pilots must use caution when flying in foggy conditions and rely on their instruments to navigate safely.
- If visibility becomes too low, pilots may need to divert to another airport or delay their flight until conditions improve.

Other Weather Phenomena Causing Reduced Surface Visibility

- In addition to fog, there are several other weather phenomena that can cause reduced surface visibility, including:-
  - **mist**,
  - **sandstorms** and **dust storms**,
  - **precipitation**, and
  - various **lithometeors**.

- **Mist**: Mist is similar to fog, but the visibility reduction is not as severe.
  - ✓ Mist is formed when moisture in the air condenses into tiny water droplets near the ground, but the droplets are **not as dense** as those in fog.
  - ✓ Mist is commonly seen in **coastal areas** and can also form over **lakes** and **rivers**.
  - ✓ It can reduce visibility to a few hundred meters and can make driving and flying more challenging.

- **Sandstorms and Dust Storms**: Sandstorms and dust storms are common in **arid** and **desert** regions.
  - ✓ They occur when **strong winds** lift sand and dust particles off the ground and carry them into the air.
  - ✓ These storms can reduce visibility to near-zero levels, making it dangerous to drive and fly.
  - ✓ They can also cause respiratory problems and damage to buildings and other structures.

- **Precipitation**: Precipitation, such as rain, drizzle, and snow, can also cause reduced visibility.
  - ✓ Rain and drizzle can create a **fine mist** that can reduce visibility to a few hundred meters.
  - ✓ Snow can create blowing snow, which can reduce visibility to near-zero levels.
  - ✓ Blowing snow occurs when **strong winds** lift snow off the ground and carry it into the air, creating a **whiteout effect**.

- **Lithometeors**: Lithometeors are weather phenomena that are composed of solid particles, such as smoke, haze, and ash.
  - ✓ **Smoke** from **wildfires** and **haze** from **pollution** can reduce visibility, making it difficult to flying aircraft.
  - ✓ **Ash** from **volcanic** eruptions can also cause reduced visibility and can be hazardous to aircraft engines.

- In summary, mist, sandstorms and dust storms, precipitation, and various lithometeors can all cause **reduced surface visibility**.
- It is important to take precautions when flying in these conditions and to rely on instruments for navigation and safety.

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Thanks !